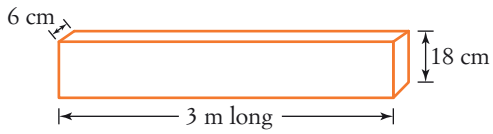
WORKSHEET

Painting the fence



The post-and-rail fence on Craig's rural property needs painting. Because he has horses, Craig has to use white paint on the fence. Complete the questions on this worksheet to determine the quantity of paint Craig will need for the job.

1 The diagram shows the dimensions of each rail on the fence.



a Express 6 cm and 18 cm in metres.

b Use a calculation to show that the total area of the front and back of each rail is 1.08 m^2.

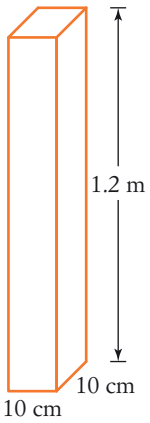
c Show that the total area of the top and the bottom of each rail is 0.36 m^2.

d Calculate the total area of the front, back, top and bottom of each rail.

e Write a sentence to explain why we don't need to include the area of the ends of the rails in the area to be painted.

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2 The diagram shows the fence posts. Each post is 1.2 m high and the cross section is a square with sides 10 cm.



a Use a calculation to show that the area of the front of each post is 0.12 m^2 and that the area of the top of the post is 0.01 m^2.

b Calculate the total area of the 5 faces of the fence posts that require painting. Ignore any overlapping area between the posts and the rails.

3 Craig's fence is 1200 m long and it has a top and a bottom rail each 3m long.

a How many fence rails are there?

b Explain why there are 401 vertical fence poles.

c Calculate the total area Craig will be painting on the posts and rails.

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4 Each litre of paint covers 13 m^2. How many litres of paint will Craig need?

5 The paint comes in 15 L containers.

a How many containers will Craig need for the job?

b Each 15 L container costs $109. Calculate the total cost of the paint.

c Craig receives a large quantity purchase discount of 25%. How much will Craig have to pay for the paint?

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Answers

1 **a** 0.06 m, 0.18 m

b $2 \times 3 \times 0.18 = 1.08$

c $2 \times 3 \times 0.06 = 0.36$

d 1.44 m^2

e They butt up to the adjacent rail.

2 **a** $0.1 \times 1.2 = 0.12$ m^2, $0.1 \times 0.1 = 0.01$ m^2

b 0.49 m^2

8 **a** 800

b You need one for each length of rail plus one for the end.

c $800 \times 1.44 + 401 \times 0.49 = 1348.49$ m^2

4 103.73 L

5 **a** 7

b $763

c $572.25

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